If the class starts at 09:05 PM IST and I have told you that I will start the class in the next 5 minutes then what's the probability that I will start the class at 09:09 PM IST (Remember, I may wait for few minutes or seconds based on the attendance in the session.) $= 0$

$$\text{Prob}(x) = \frac{\#\text{ of favourable outcomes}}{\text{Total no. of outcome}}$$ (all observation in the sample space)

Toss of two coins

{HH, HT, TH, TT}

P(HH)

09:05:01

{09:05, 09:06 . . . . . 09:10}

09:05:01:02

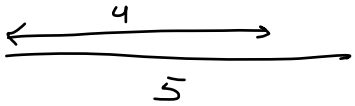
$$= \frac{09:09}{\infty} = 0 \quad \left\{ \frac{1}{\infty} \right\}$$

Continuous R.V

intervals =

$$P(09:05 - 09:09) = \frac{4 \text{ mins}}{5 \text{ mins}} = \frac{4}{5}$$ density

$$P(09:05 - 09:07) = \frac{2 \text{ min}}{5 \text{ min}} = \frac{2}{5}$$



Introduction

Recruiting and retaining drivers is seen by industry watchers as a tough battle for Ola. Churn among drivers is high and it’s very easy for drivers to stop working for the service on the fly or jump to Uber depending on the rates.

As the companies get bigger, the high churn could become a bigger problem. To find new drivers, Ola is casting a wide net, including people who don’t have cars for jobs. But this acquisition is really costly. Losing drivers frequently impacts the morale of the organization and acquiring new drivers is more expensive than retaining existing ones.

You are working as a data scientist with the Analytics Department of Ola, focused on driver team attrition. You are provided with the monthly information for a segment of drivers for 2019 and 2020 and tasked to predict whether a driver will be leaving the company or not based on their attributes like

Problem statement

(papu)
Driver

(OLA)

churn

chennai)

(Airport

OLA

↓

uber (Ride found)

OLA

UBER

MERU

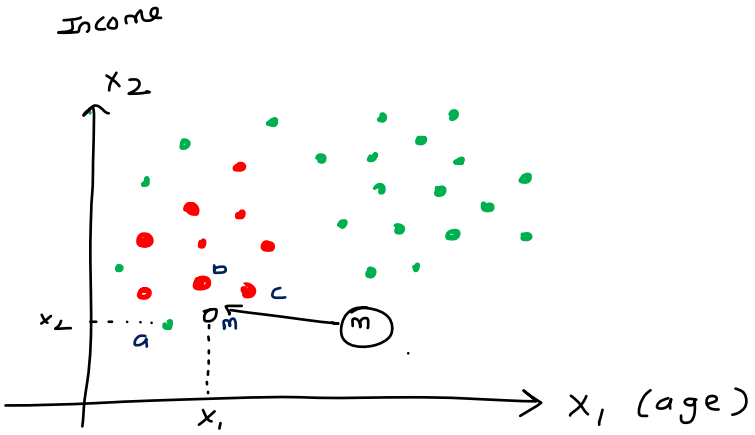
Fast track

1. Less available cabs at a certain location
2. customer churn
3. Loss

Either Retain
or
Replacement

age	income	gender
		NA

cat



$K = 3$

a → green (F)
b → red (m)
c → red (m)

$\frac{a+b+c}{3}$ → missing

Class Imbalance :

y
- 1 : 20%
- 0 : 80%

} Imbalanced data

y
- 1 : 50%
- 0 : 50%

} Balanced data

1 — 68%
0 — 32%

Techniques
- Random oversampling
- Random undersampling
- SMOTE
- Adsyn
- Class-weight

}

Random oversampling

indep	y
0	1
1	0
2	0
3	0
4	1
5	0
6	1
7	0
8	0
9	1

y < 1 : 1 %, 0 : 99 % $\xrightarrow{Ro}$ B → (m)

Randomly increase / duplicate the no. of rows for minority class

indep	y
0	1
0	1
1	0
2	0
3	0
4	1
5	0
6	1
7	0
8	0
9	1
9	1

= 1000

y < 1 : 50 %, 0 : 50 %

1. No new information is getting added
2. overfitting

y < 1 : 40 %, 0 : 60 %

SMOTE

Synthetic minority class oversampling technique

we generated artifical rows for the minority class to balance it out

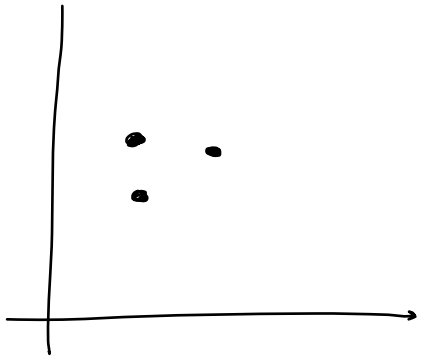
x_1	x_2	x_3	y
5	2	9	0
7	3	5	1
6	2	5.1	1

Random oversampling

artifical row

y < 1 : 1 %, 0 : 99 %

$\xrightarrow{\text{SMOTE}}$



(will discuss in end)

we assign more weight to ~~more~~ minority class and less weight to majority class

Class-weight → all sklearn classification algo

Logistic Regression

Cost function

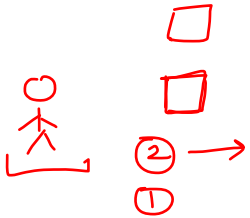
$$\begin{cases} \log(y(x)) & \text{if } y = 1 \\ 1 - \log(y(x)) & \text{if } y = 0 \end{cases}$$

↓

y < 1 : 20% , 0 : 80%

cost-function

$$\begin{cases} 0.8 \log(y(x)) & \text{if } y = 1 \\ 0.2(1 - \log(y(x))) & \text{if } y = 0 \end{cases}$$



↓

class-weight = 'Balanced'

{1 : 0.8 , 0 : 0.2}

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\uparrow R \propto \frac{1}{FN \downarrow}$$

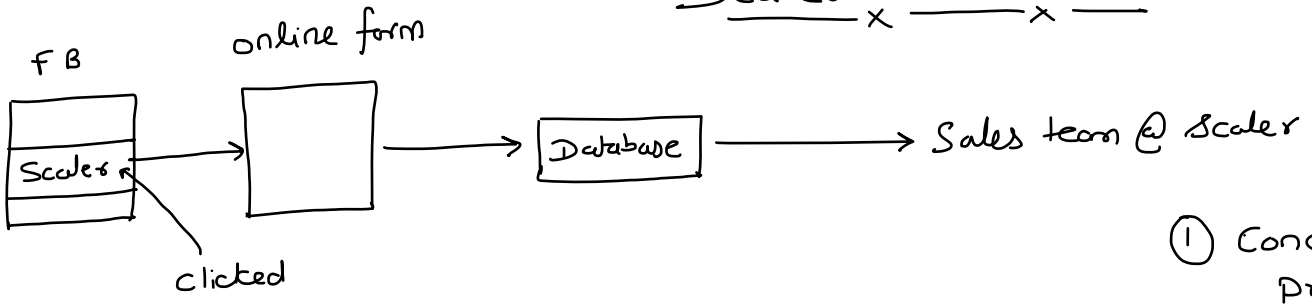
$$\text{precision} = \frac{TP}{TP + FP}$$

$$\uparrow P \propto \frac{1}{FP \downarrow}$$

$$\text{f1-Score} = \frac{2 \times P \times R}{P + R}$$

will only lead to high value when both P & R is high

Scaler Business case



1. Concrete data to explain prospect how this course will benifit them

Data → Students currently enrolled in various programs.

Current Student

	company	email	year of Joining	current CTC	Position	CTC updated	CTC_Level	Pos-Level	Joing-Level	Yr. of exp
S1 →	Infosys			5L			3			
S2 →	amazon									
	google									

CTC up
−
yr. of Joing

[Data pre-processing]

company	25% (1)	75% (3 → 2)	100% (1)
Infosys	2.5 L	6L	15L
amazon			
google			

Company	Post	25%	75%	100%
Infosys	S.E	-	.	.
Infosys	Support	-	.	.
Infosys	Testing	.	.	.

$x \geq 100\% \rightarrow 1$

$75\% \leq x < 100\% \rightarrow 2$

$25\% \leq x < 75\% \rightarrow 3$

$x < 25\% \rightarrow 4$

Company	Post	yr. of joing	25	75	100

Data → Algo (K-mean) → 1, 2, 3

1 → CTC (3,4) ; (3,4) ; (3,4) ; Low ; (0,1)

Sales ← New student

$y = 1$

Infosys

pos = S.E

y.o.j = 2019

C.S = 4 LPA